

Question#1:

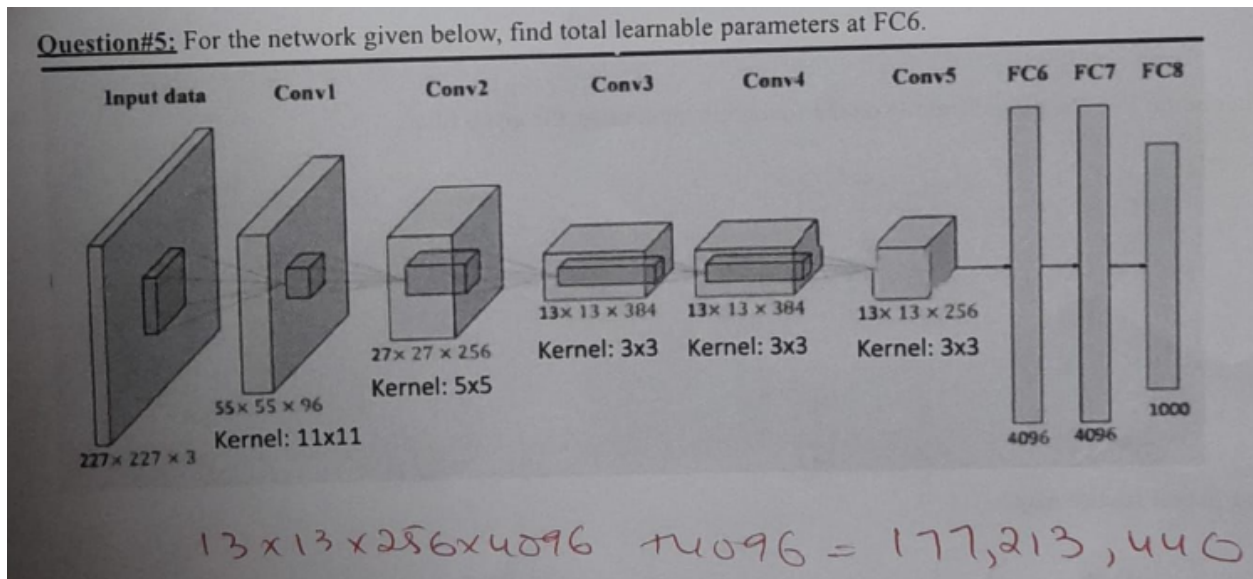
Layer 1 (Conv2D): Input Shape (32, 32, 3), 8 Filters, Kernel Size (3, 3), Stride 1, no Padding.

Layer 2 (Conv2D): Input comes from Layer 1, 16 Filters, Kernel Size (3, 3), Stride 1, no Padding.

Calculate the total number of learnable parameters (weights + biases) specifically for Layer 2.

$$3 \times 3 \times 8 \times 16 + 16 = 1168$$

Question#5: For the network given below, find total learnable parameters at FC6.



Part a: Output Dimensions

Compute the output size after each layer:

- After first convolution
- After first pooling
- After second convolution
- After second pooling

Part b: Number of Parameters

Compute total trainable parameters in:

- First convolution layer
- Second convolution layer